



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

REPORT PROJECT 1: PROBABILITY & STATISTICAL DATA ANALYSIS (SI1143-03)

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1. Introduction

Our project is to investigate the impact of video games on student academic performance. In the modern era, playing video games has become one of the most popular leisure activities in recent times, especially among students, who often turn to it to relax and ease the overwhelming obligations of academic work. Indeed, this trend has been associated with the desire to mitigate the rising pressure and stress associated with the multitude of assignments and course requirements imposed by their teachers. However, this trend has multifarious effects, with both positive and negative repercussions depending on the students' ability to control their gaming habits relative to their academic commitments.

Unfortunately, many students can become victims of video game addiction and thus have been observed to invest huge sums of time in virtual worlds at the cost of critical parts of their lives, including messing up sleep schedules. Such a tendency can result in a decline in their focus in lectures and, subsequently, a deterioration of their grades, manifesting in a negative impact on their CGPAs.

So that's why, In this project, we will question UTM students about how the video game affects their academic performance. We want to find out how many students play video games in their leisure time, how many hours they spend on them, if the video game gives them a bad habit, and whether the student needs to put away the game habit in order to succeed in academics. This will help us identify whether video games can affect academic performance.

The questionnaire is distributed to students at UTM through the use of Telegram and WhatsApp; this results in responses from 65 respondents. Our hypothesis is that students who spend a lot of time gaming will likely have a lower CGPA than those with lower gaming activities.

2. Data Collection

No	Question	Answer	Level of Measurement
1	Gender	• Male • Female	Nominal
2	Age	• Age	Ratio
3	Ethnics	• Malay • Chinese • Indian • Other	Nominal
4	Academic year	• Year 1 • Year 2 • Year 3 • Year 4	Nominal
5	Faculty	• Faculty of Civil Engineering (FKA) • Faculty of Mechanical Engineering (FKM) • Faculty of Chemical & Energy Engineering (FKT) • Faculty of Electrical Engineering (FKE) • Faculty of Computing (FC) • Faculty of Built Environment and Surveying (FABU) • Faculty of Science (FS) • Faculty of Social Sciences & Humanities (FSSH) • Faculty of Management (FM)	Nominal
6	Current CGPA	• CGPA	Ratio
7	Do you play video games?	• Occasionally • Sometimes • Often • Always	Ordinal
8	How many hours do you play a game per day ?	• Hours	Ratio
9	Does video games affect the students' academic performance?	• Strongly agree • Agree • Disagree • Strongly Disagree	Ordinal

No	Question	Answer	Level of Measurement
10	Can video games develop bad habits in students ?	• Strongly agree • Agree • Disagree • Strongly Disagree	Ordinal
11	What are the bad habits that can be developed by playing a video game?	• Inconsistent sleep schedules • Unfocus on lecture • Obsession on game • Neglect the assignment • Other	Nominal
12	Do you agree that video games have their own benefits ?	• Strongly agree • Agree • Disagree • Strongly Disagree	Ordinal
13	What are the benefits of playing games?	• Reduce stress level • Enhance critical thinking • Enhance problem-solving skill • Improved communication skill • Other	Nominal
14	In your opinion, does the student need to put away the game habit in order to succeed academically?	• Opinion (Yes/No with elaboration)	Nominal

3. Data Analysis

3.1 BIOGRAPHY OF THE RESPONDENT

3.1.1 Respondent's Gender

Gender	Frequency	Percentage
Male	41	63%
Female	24	37%
TOTAL	65	100%

Table 3.1.1.1 Frequency and Percentage of Gender of Respondent

Genders of Respondents

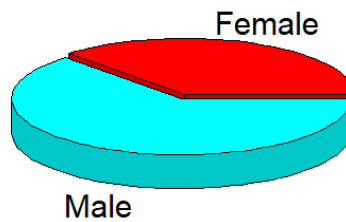


Figure 3.1.1.2 Pie Chart of Gender of Respondent

As indicated in Table 3.1.1.1 and Figure 3.1.1.2., which are the frequency table and pie chart of respondent gender, the frequency of male are 41 and females are 24 . As a result, the relative frequency and proportion for male are 0.63 and 63% while for females are 0.37 and 37%. There are 65 respondents of each gender. In addition, the pie chart of responder gender illustrates that red means female and blue means male.

3.1.2 Respondent's Age

Age	Frequency
19	3
20	41
21	7
22	7
23	6
24	1
TOTAL	65

Table 3.1.2.1 Frequency Distribution of Age of Respondent

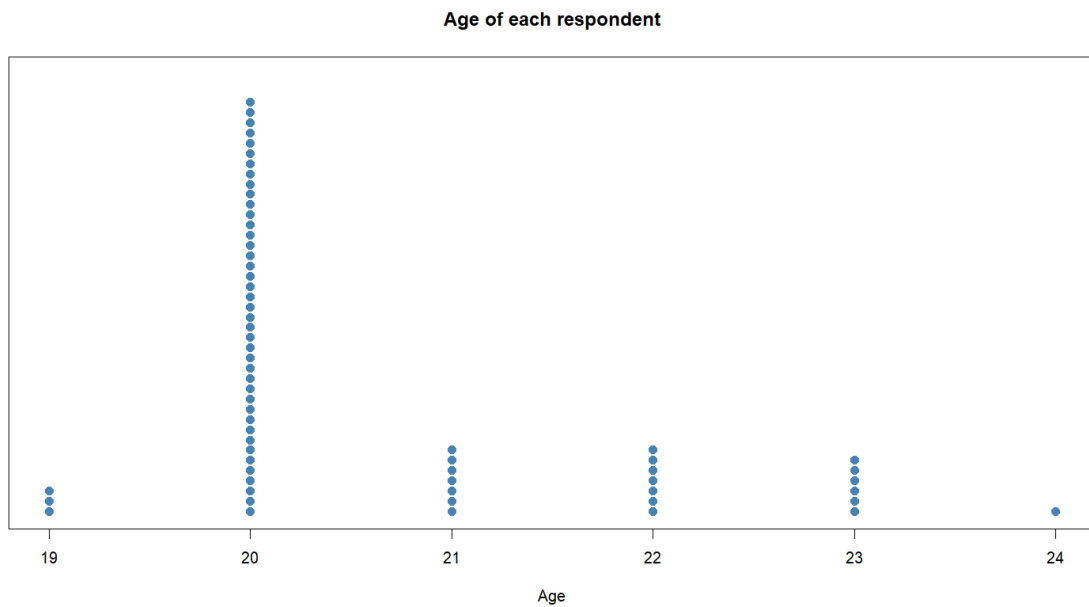


Figure 3.1.2.2 Dot Plot of Age of respondent

Measure of Central Tendency	Value
Mean	20.6153
Median	20
Mode	20

Table 3.1.2.3 Measure of Central Tendency of Age of Respondents

The age distribution of respondents who have completed our questionnaire falls between the ages of 19 and 24, as can be seen in Table 3.1.2.1 and Figure 3.1.2.2. Twenty-year-olds make up the majority of responders, with a frequency of 41. Next is the students of age 21 and 22 with the frequency of 7, followed by 6 responders who are 23 years old. There are also 3 19-year-old and 1 24-year old respondents. Furthermore, Table 3.2.3 makes clear that the majority of the respondents who provided the data were in the age range of 20 as our median and mode both are 20, and mean being 20.6153.

3.1.3 Respondent's Ethnicities

Ethnicities	Frequency
Malay	41
Chinese	17
Indian	5
Indonesian	1
Egyptian	1
TOTAL	65

Table 3.1.3.1 Frequency of Ethnicities respondent

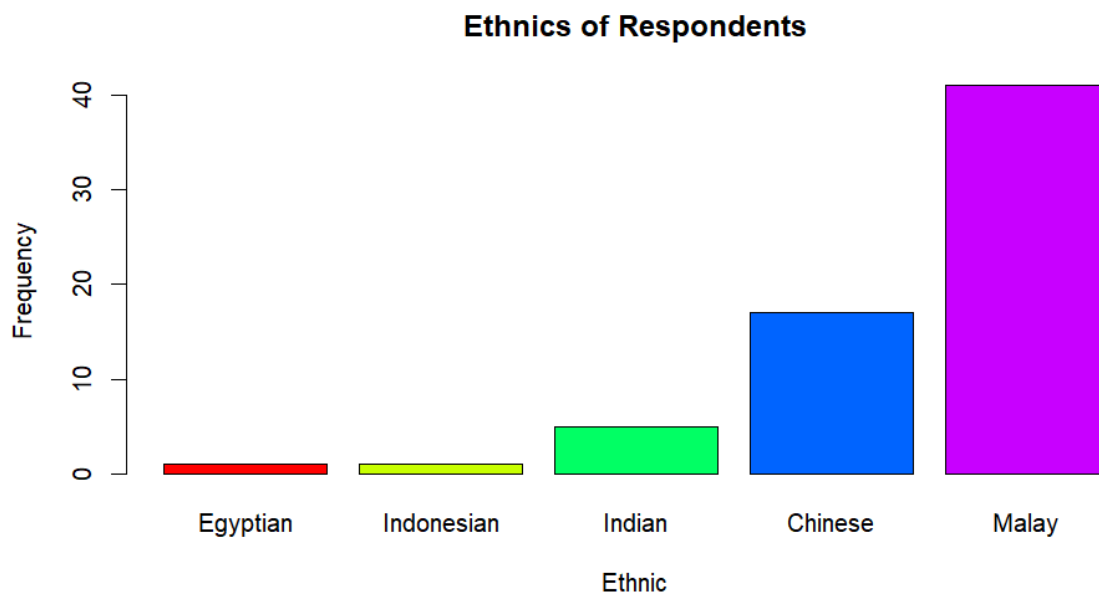


Figure 3.1.3.2 Bar chart of Ethnicities respondent

The distribution of ethnicities among respondents in a UTM is shown in Table 3.1.3.1 and Figure 3.1.3.2. Among them, Malay people make up the largest ethnic group with a frequency of 41, followed by Chinese people with a frequency of 17, and Indian people with a frequency of 5. There is a smaller representation of Indonesian and Egyptian ethnicities, each accounting for 1 frequency.

3.1.4 Respondent's Academic Year

Academic Year	Frequency
Year 1	40
Year 2	10
Year 3	10
Year 4	5
TOTAL	65

Table 3.1.4.1 frequency distribution of Academic year of respondent

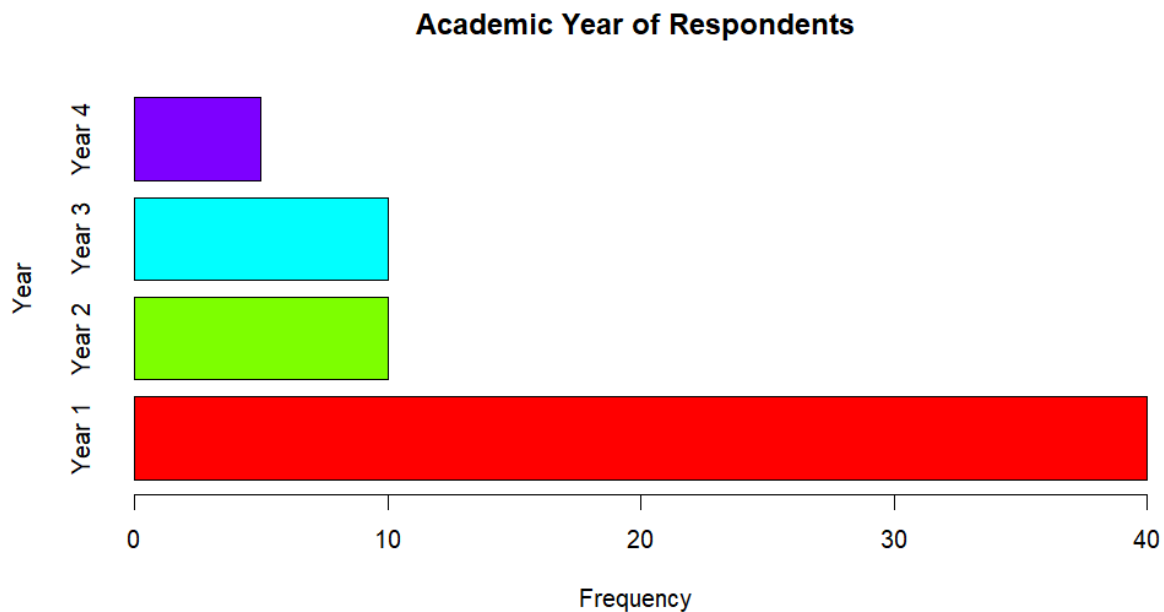


Figure 3.1.4.2 Bar Chart of Academic Year of Respondents

Table 3.1.4.1 and Figure 3.1.4.2 shows the distribution of students across different years in a UTM, based on a sample population of 65 students. Most of the students, 40 in total, represent the first academic year, followed by 10 students in the second and third years. There is also small representation in the fourth year, with 5 persons in the sample.

3.1.5 Respondent's Faculty

Faculty	Frequency	Percentage (%)
Faculty of Civil Engineering (FKA)	3	4.6
Faculty of Mechanical Engineering (FKM)	6	9.2
Faculty of Chemical & Energy Engineering (FKT)	6	9.2
Faculty of Electrical Engineering (FKE)	4	6.2
Faculty of Computing (FC)	33	50.8
Faculty of Built Environment and Surveying (FABU)	2	3.1
Faculty of Science (FS)	5	7.7
Faculty of Social Sciences & Humanities (FSSH)	1	1.5
Faculty of Management (FM)	5	7.7

Figure 3.1.5.1 Frequency and percentage of respondent's faculty

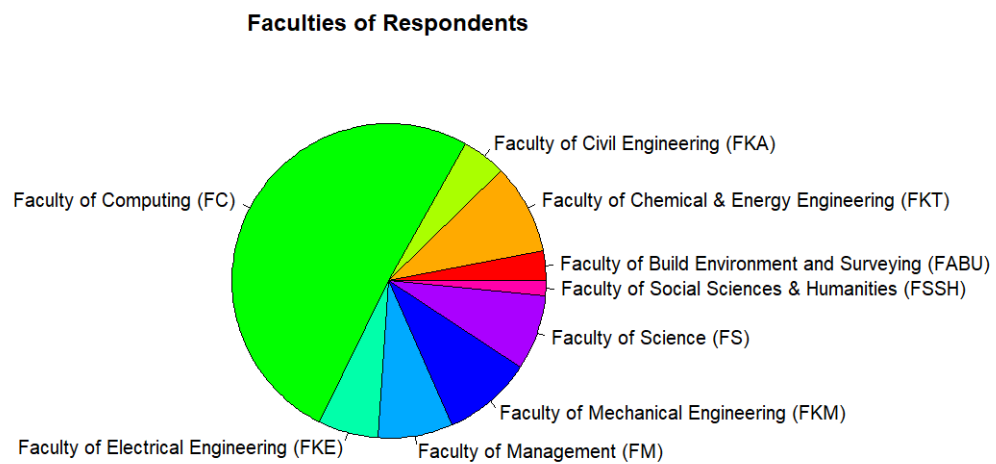


Figure 3.1.5.2 Pie chart of respondent's faculty

Figure 3.1.5.1 and figure 3.1.5.2 show the department that might be the respondent's department or specifically student's faculty. The most respondents are from the Faculty of Computing (FC) which is 33 respondents. Followed by both Faculty of Mechanical Engineering (FKM) and Faculty of Chemical & Energy Engineering (FKT) which has 6 respondents for both faculties. Next is Faculty of Science (FS) and Faculty of Management (FM) which has 5 respondents for both. After that is the Faculty of Electrical Engineering (FKE) with 4 respondents. Faculty of Civil Engineering (FKA) is the next with 3 respondents. 2 respondents from Faculty of Built Environment and Surveying (FABU) and the least respondents is from Faculty of Social Sciences & Humanities (FSSH) with 1 respondent only.

3.1.6 Respondent's Cumulative Grade Point Average

CGPA	Frequency	Cumulative frequency
1.80 - 2.00	1	1
2.00 - 2.20	0	1
2.20 - 2.40	0	1
2.40 - 2.60	0	1
2.60 - 2.80	2	3
2.80 - 3.00	2	5
3.00 - 3.20	1	6
3.20 - 3.40	2	8
3.40 - 3.60	6	14
3.60 - 3.80	16	30
3.80 - 4.00	35	65
TOTAL	65	65

Table 3.1.6.1 Frequency of Respondent's Cumulative Grade Point Average

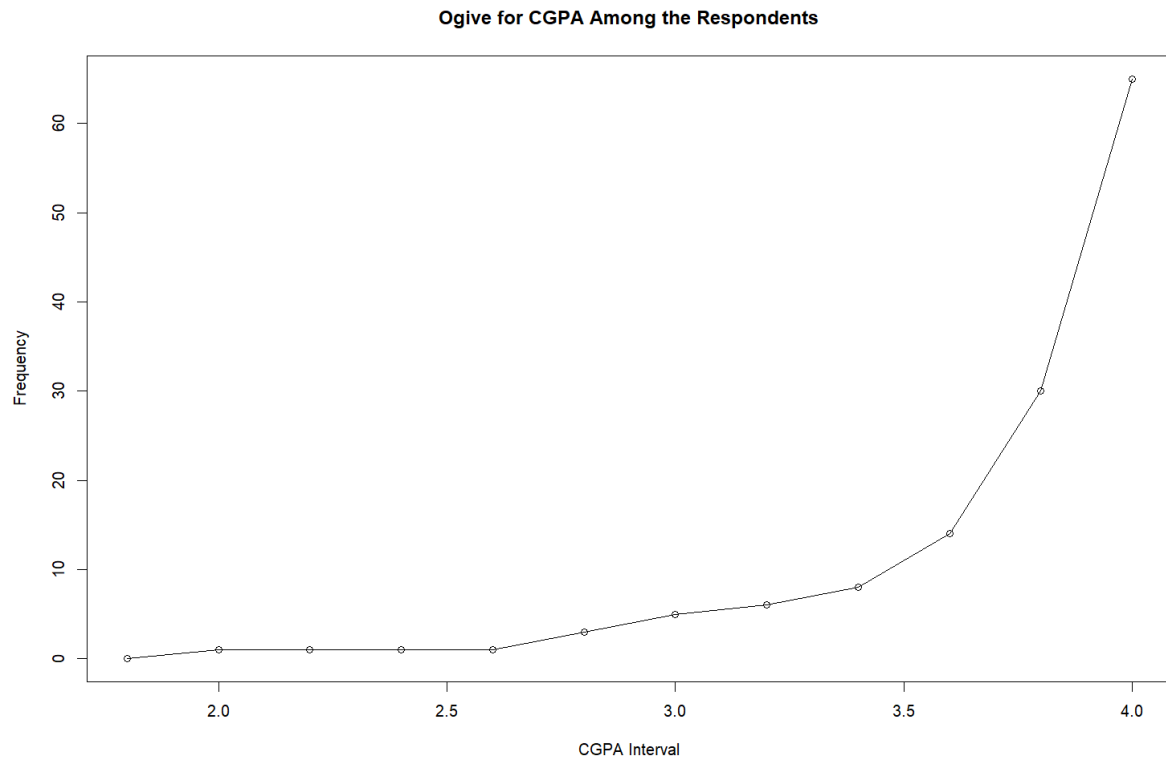


Figure 3.1.6.2 Ogive for CGPA among the respondents

Table 3.6.1 and Figure 3.6.2 state the frequency distribution of the cumulative grade point average (CGPA) of students in an academic setting with 65 students. The CGPA is in the range of 1.80 to 4.00. 35 students have a CGPA in the 3.80 to 4.00 range and, thus, represent the highest performance range. There are 16 students in the range of 3.60 to 3.80, 6 students in the range of 3.40 to 3.60, 2 students in the range of 3.20 to 3.40, and 2 students in the range of 3.20 to 3.40, and, lastly, 2 students in the range of 2.60 to 2.80, and 1 student in the range of 1.80 to 2.00, and in the range of 3.00 to 3.20. There is one student each in the 2.00 to 2.20 range and in the 2.20 to 2.40 range. lastly there are no students who have a CGPA between 2.00 and 2.20, nor between 2.20 and 2.40.

3.2 RECONDENT OPINION ON VIDEO GAME (GENERAL)

3.2.1 Do you play video games?

How often	Frequency	Percentage (%)
Always	12	18.5
Often	17	26.2
Sometimes	27	41.5
Occasionally	9	13.8

Figure 3.2.1.1 Frequency and percentage of respondent on how often play games

How Often the Respondants Play Games

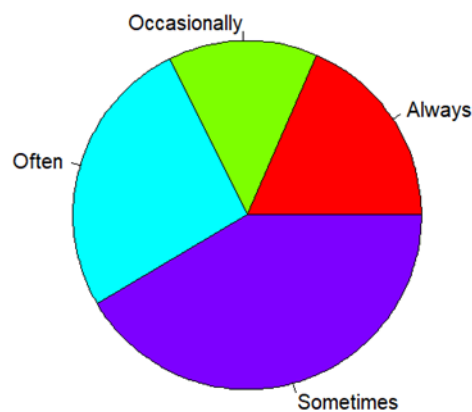


Figure 3.2.1.2 Pie chart of respondents on how often play game

Figure 3.2.1.1 and figure 3.2.1.2 show how often respondents play games. Most of the respondents play games sometimes with 27 respondents on it. But respondents often play games with 17 respondents. Besides, respondents that always play games just 12 respondents. At the end 9 respondents occasionally play games. So, respondents play games sometimes in their life as university students.

3.2.2 How many hours do you play a game per day ?

Hours	Frequency	Cumulative frequency
0	3	3
0.5	4	7
1	12	19
2	11	30
3	3	33
4	9	42
5	5	47
6	5	52
7	4	56
8	3	59
9	4	63
10	2	65
TOTAL	65	65

Table 3.2.2.1 Frequency of hours students spend playing video games

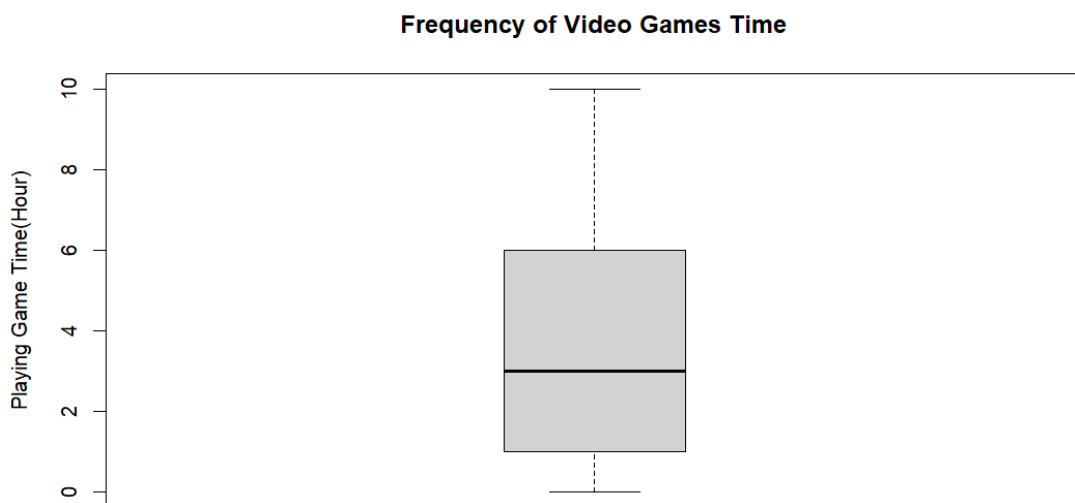


Figure 3.2.2.2 BoxPlot of frequency of hours students spend playing video games

Quartile	Hours
0.25	1
0.50	3
0.75	6

Table 3.2.2.3 Measure of Quartile for boxplot of frequency of hours students spend playing video games

Table 3.2.2 displays the frequency distribution of the amount of hours a respondent spends their time on video games. The range of hours is from 0 to 10. There are 3 people who report not having spent any time on the video game, and 4 people report having spent 30 minutes on it. Then, within the 1 to 4 hour range, the frequency rises very gradually: 12 people spend 1 hour, 11 people spend 2 hours, and 3 people each spend 3 and 4 hours on the video game. It seems that quite a few people in the sample spent relatively little time on the activity. As the hours increase, the frequency remains relatively constant: 9 people spend 4 hours on video games, 5 spend 5 and 6 hours, and 4 spend 7 and 8 hours, respectively. On the high side there are 2 people who each put 9 and 10 hours into the activity.

Next in figure 3.2.2.2, we can see that the box is skewed to the bottom, which indicates that most of the students are playing video games for 1 to 6 hours a day. In table 3.2.2.3, our first quartile lies at 1 hour, our second quartile lies at 3 hours, and our third quartile lies at 6 hours. This data tells us that most of the students play video games in their leisure time.

3.2.3 Does video games affect the students' academic performance?

Effect the students' academic performance	Frequency	Percentage (%)
Strongly agree	12	18.5
Agree	32	49.2
Disagree	18	27.7
Strongly disagree	3	4.6

Table 3.2.3.1 Frequency and percentage of respondent opinion about game effect academic or not

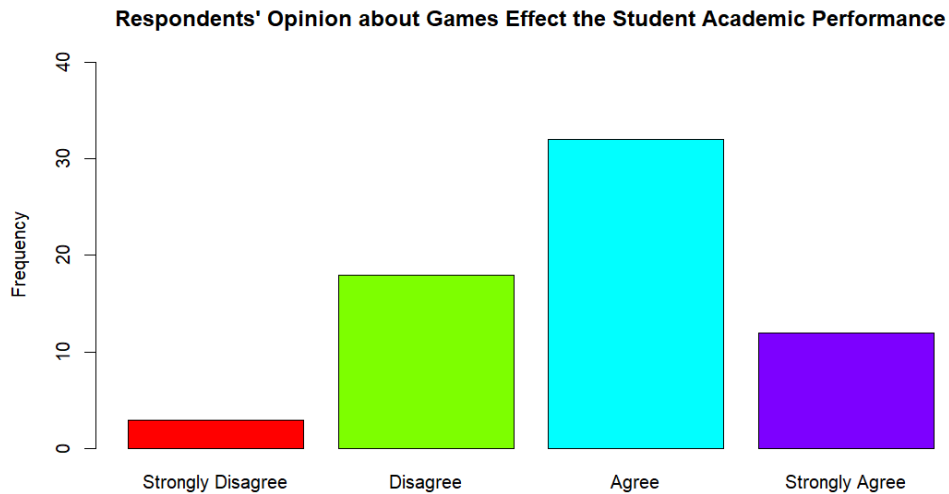


Figure 3.2.3.2 Bar chart of respondent opinion about game effect or not

Table 3.2.3.1 and figure 3.2.3.2 show respondents' opinions about games affecting student academic performance. Most respondents agree with 32 respondents. Meanwhile, only 18 respondents disagree. Strongly agree respondents are 12 respondents and 3 respondents strongly disagree. So, most respondents agree that playing games can affect student academic performance.

3.2.3 Does video games develop bad habits in students ?

Develop a bad habits	Frequency	Percentage (%)
Strongly agree	20	30.8
Agree	27	41.5
Disagree	13	20
Strongly disagree	5	7.7

Table 3.2.3.1 Frequency and percentage of respondent opinion about bad habits play video game

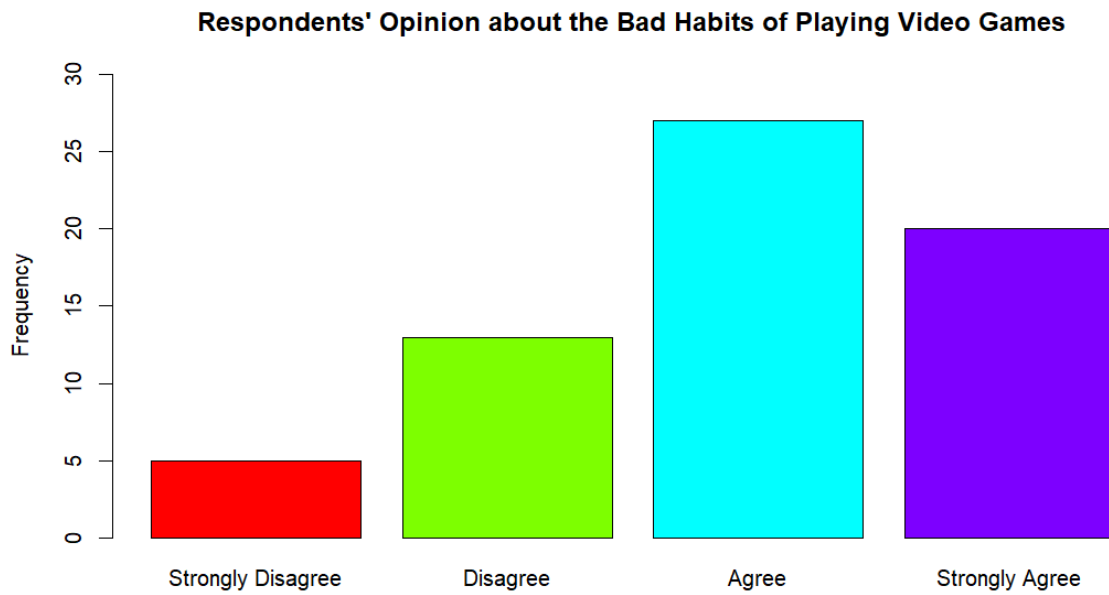


Figure 3.2.3.2 Bar chart of respondent opinion about bad habits play video game

Table 3.2.3.1 and figure 3.2.3.2 show respondents' opinions about the bad habits of playing video games. With 27 respondents, they agree on it. But respondents that disagree are about 13 respondents. Meanwhile, respondents strongly agree with 20 respondents which makes respondents that strongly disagree is the least with 5 respondents. So, most respondents agree that by playing video games it can create bad habits among students.

3.2.4 What are the bad habits that can be developed by playing a video game?

Bad habits	Frequency
Inconsistent sleep schedules	46
Unfocus on lecture	24
Obsession on game	46
Neglect the assignment	29
Swearing	1
Does Not have any bad habit	1

Table 3.2.4.1 Frequency of bad habits among respondents

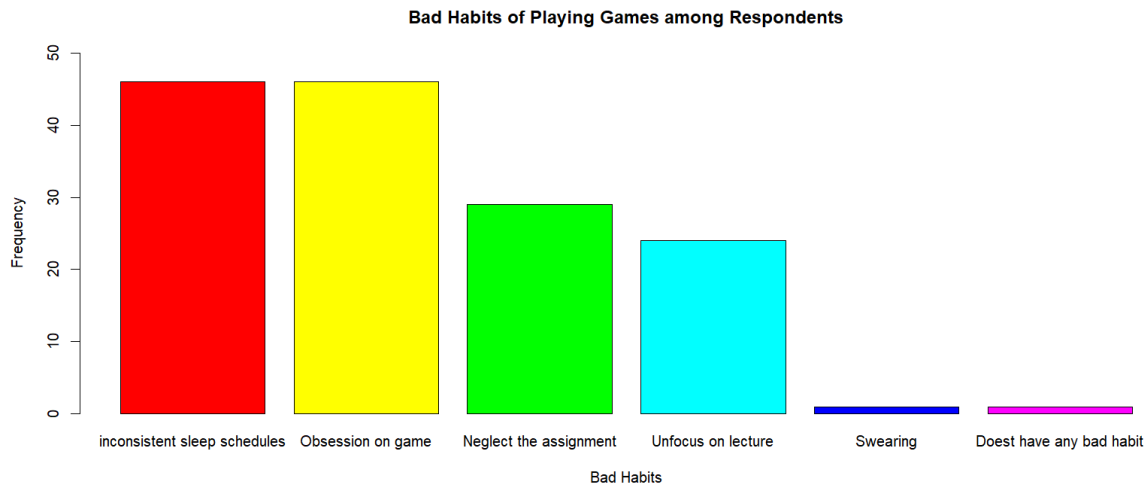


Figure 3.2.4.2 Bar chart of bad habits among respondents

Table 3.2.4.1 and figure 3.2.4.2 show bad habits of playing games among respondents. Out of 65 respondents, 46 respondents for both inconsistent sleep schedules and obsession with games. Followed by Neglect the assignment with 29 respondents. Next with 24 respondents is on unfocus on lecture. Meanwhile, at least they swear with 1 respondent only and 1 respondent that thinks that video games don't have any bad habits. So, most respondents' bad habits of playing video are inconsistent sleep schedules and obsession with games.

3.2.5 Do you agree that video games have their own benefits ?

Have benefits	Frequency	Percentage (%)
Strongly agree	25	38.5
Agree	34	52.3
Disagree	6	9.2
Strongly disagree	0	0

Table 3.2.5.1 Frequency and percentage of respondents opinion about benefits

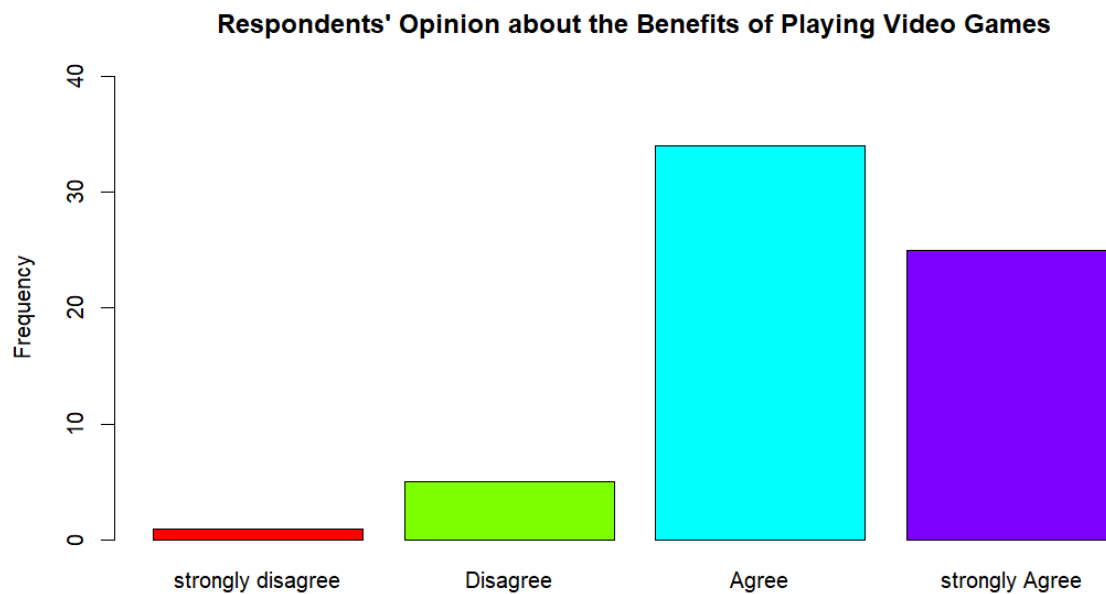


Figure 3.2.5.2 Bar chart of respondents opinion about benefits

Table 3.2.5.1 and figure 3.2.5.2 show respondents' opinions about the benefits of playing video games. As the agreed respondents are the most with 34 respondents followed by 25 respondents that strongly agree. Leave only respondents that disagree with 6 respondents. 0 respondents that strongly disagree. So, despite most respondents agreeing on bad habits of playing video games, they agree that playing video games also gives benefits.

3.2.6 What are the benefits of playing games?

Benefits	Frequency
Reduce stress level	48
Enhance critical thinking	36
Enhance problem solving skill	33
Improved communication skill	28
Strengthen friendship	1
Does Not have any benefits	3

Table 3.2.6.1 Frequency of benefits among respondents

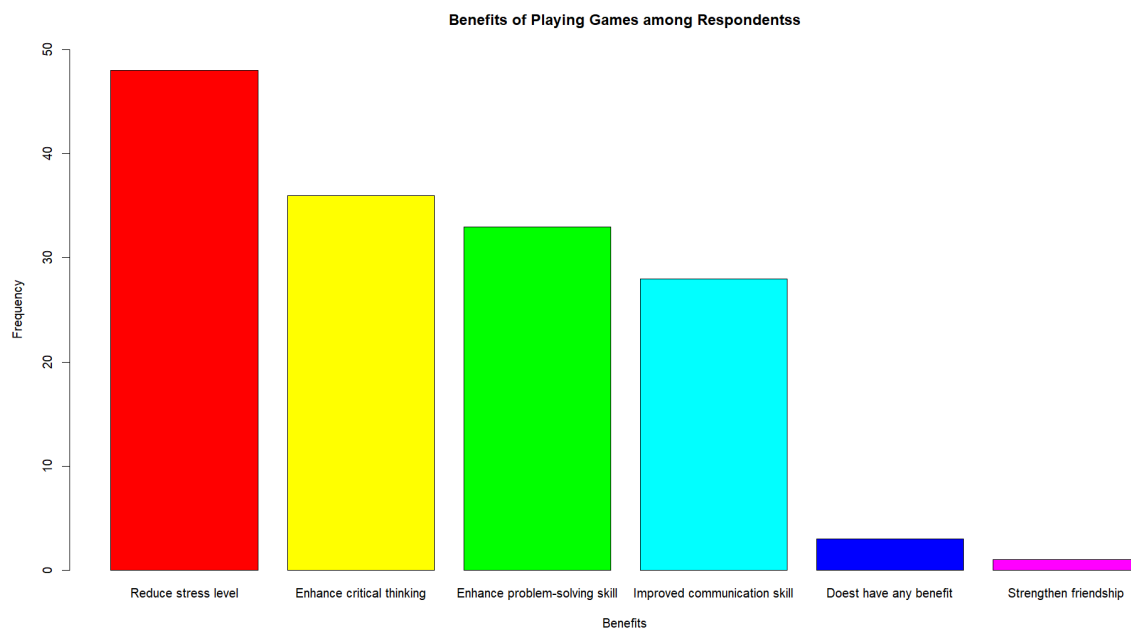


Figure 3.2.6.2 Bar chart of benefits among respondents

Table 3.2.6.1 and figure 3.2.6.2 are about benefits of playing games among respondents. With 48 respondents, they vote to reduce stress levels. Followed by enhanced critical thinking with 36 respondents. Next, with 33 respondents, is to enhance problem solving skills, 28 respondents, is to improve communication skills and 1 respondents to strengthen friendship. However, 3 respondents agree that there are no benefits for playing video games. So, most respondents are playing games to reduce stress levels.

3.2.7 In your opinion, does the student need to put away the game habit in order to succeed academically?

Put away game habits	Frequency
Yes	21
No	38
Depend	5
Don't know	1

Figure 3.2.7.1 Frequency of respondents opinion on putting away game habit

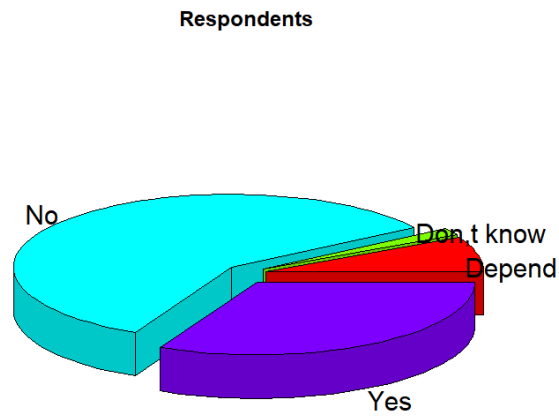


Figure 3.2.7.2 Pie chart of respondents opinion on putting away game habit

Figure 3.2.7.1 and figure 3.2.7.2 show respondents' opinions about putting away the game habit in order to succeed in academics. With the most with 38 respondents, respondents say no. followed by respondents that said yes with 21 respondents. But, 5 respondents said it depends. Meanwhile, only 1 respondent is clueless and says they don't know. So, most respondents said that there is no need to put game habits in order to succeed.

4. Conclusion

Our research had concluded with the 65 respondents throughout all the faculty in the Universiti Teknologi Malaysia (UTM), consisting of 41 males and 24 females. Based on our survey, the high frequency which is 27 of 65 students are sometimes playing games during their studies. They usually play video games around 4 - 6 hours per day. In this case, game addiction has a significant effect on the sleeping pattern of students. It has been found that students who are addicted to playing video games tend to have poorer sleep quality and experience difficulty in falling asleep and waking up at appropriate times. Like in this survey, around 46 respondents choose the Inconsistent sleep schedules as the answer for what are the bad habits that can be developed by playing video game questions. While video games can bring consequences towards the students, most of the respondents agree that video games can also have their own benefits. There are 59 respondents who agree in this questionnaire. These days, a lot of video games demand players to go through challenging riddles and issues, which helps develop critical thinking abilities. In general, a lot of students experience personal stress from time to time, particularly from assignments and exercise. Which is why video games can help to maintain their mental health level. This will indirectly help them overall their studies to achieve excellent performance.

To sum up our research, we would like to bring forward the vast amount of knowledge we gained from conducting our survey. We believe the conduction of this research will assist us in our future in terms of both technical skills and communication skills, allowing us to be an efficient worker in future companies. We would like to give credit to our lecturer, Dr. Nor Haizan binti Mohamed Radzi for guiding us to our project and also to our 65 respondents who kindly helped us with their participation to complete our research.